Derivable Judgement

Theory Notes: Statistical Concepts Explained in Simple English

This document explains the main statistics ideas used in the project. Each topic has a simple definition, a formula or rule, and an example.

Quick table: which test to use?

Test	Use when	Data type
z-test	Large sample, known sigma	Numbers (means)
t-test	Small sample, unknown sigma	Numbers (means)
Chi-square	Check link between groups	Categories
ANOVA	Compare 3+ group means	Numbers + groups

1. Inferential Statistics

Inferential statistics means using data from a small sample to make conclusions about a bigger population. We cannot measure every person, so we study a sample and then judge the whole group.

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Example: Checking 1000 people to guess the diabetes rate of a whole city.
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2. Hypothesis Testing and Its Components

Hypothesis testing is a method to check a claim using data. Main parts: **Null hypothesis (H0)** - there is no effect. **Alternative hypothesis (H1)** - there is an effect. **Significance level (alpha)** - the risk we accept of being wrong, usually 0.05. **Test statistic** - a number calculated from data. **Decision** - reject or fail to reject H0.

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Example: H0: Smoking has no effect on diabetes. H1: Smoking affects diabetes.
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3. Confidence Interval and Critical Value

A **confidence interval (CI)** is a range that likely contains the true population value. A 95% CI means that if we repeat the study many times, about 95% of the intervals will contain the true value. The **critical value** is the cut-off point of the test statistic. If our statistic goes beyond it, we reject H0.

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Formula: CI = mean +/- critical value x (standard deviation / square root of n)
```

4. p-value

The p-value is the chance of getting our result, or a more extreme one, when H0 is true. A small p-value (less than alpha) means the data does not agree with H0, so we reject H0. A large p-value means we fail to reject H0.

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Rule: p < 0.05 -> Reject H0. p >= 0.05 -> Fail to reject H0.
```

5. Type I and Type II Errors

Type I error (false positive): we reject H0 when H0 is true. Its chance is alpha. **Type II error** (false negative): we fail to reject H0 when H0 is false. Its chance is beta. Power of a test = 1 - beta. Lowering one error usually raises the other.

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Example: Type I = saying smoking causes diabetes when it does not. Type II =
missing a real effect.
```

6. z-test

The z-test compares a sample mean with a known value, or compares two means, when the sample is large (n > 30) and the population standard deviation is known. It uses the normal distribution.

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Formula: z = (sample mean - population mean) / (sigma / square root of n)
```

7. t-test

The t-test is used when the sample is small or the population standard deviation is unknown. It uses the t-distribution, which has heavier tails. Types: one-sample, two-sample (independent) and paired.

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Formula: t = (sample mean - population mean) / (s / square root of n)
```

8. Chi-square Test

The chi-square test checks if two categorical variables are related (test of independence), or if observed counts match expected counts (goodness of fit).

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Formula: chi-square = sum of (Observed - Expected)^2 / Expected
```

9. ANOVA (Analysis of Variance)

ANOVA checks if the means of three or more groups are different. It compares the variation between groups with the variation inside groups using the F-statistic. A large F means the group means are not all equal.

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Formula: F = variance between groups / variance within groups
```

10. Covariance

Covariance shows whether two variables move in the same direction. Positive: both rise together. Negative: one rises while the other falls. Its value depends on the units, so it is hard to compare.

```
Formula: Cov(X,Y) = sum of (X - mean X)(Y - mean Y) / (n - 1)
```

11. Correlation

Correlation is covariance scaled to be between -1 and +1. It shows the strength and direction of a linear link. +1 = perfect positive, 0 = no linear link, -1 = perfect negative. Correlation does not prove cause.

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Formula: r = Cov(X,Y) / (std X x std Y)
```